

TIPE DATA DI SQL SERVER 2012

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DATATYPE	MIN	MAX	STORAGE	TYPE	NOTES
Bigint	-2 ⁶³	2 ⁶³ -1	8 bytes	Exact	
Int	-2,147,483,648	2,147,483,647	4 bytes	Exact	
Smallint	-32,768	32,767	2 bytes	Exact	
Tinyint	0	255	1 bytes	Exact	
Bit	0	1	1 to 8 bit columns in the same table requires a total of 1 byte, 9 to 16 bits = 2 bytes, etc...	Exact	
Decimal	-10 ³⁸ +1	10 ³⁸ -1	Precision 1-9 = 5 bytes, precision 10-19 = 9 bytes, precision 20-28 = 13 bytes, precision 29-38 = 17 bytes	Exact	The Decimal and the Numeric data type is exactly the same. Precision is the total number of digits. Scale is the number of decimals. For both the minimum is 1 and the maximum is 38.
Numeric	same as Decimal	same as Decimal	same as Decimal	Exact	
Money	-2 ⁶³ / 10000	2 ⁶³ -1 / 10000	8 bytes	Exact	
Smallmoney	-214,748.3648	214,748.3647	4 bytes	Exact	
Float	-1.79E + 308	1.79E + 308	4 bytes when precision is less than 25 and 8 bytes when precision is 25 through 53	Approx	Precision is specified from 1 to 53.
Real	-3.40E + 38	3.40E + 38	4 bytes	Approx	Precision is fixed to 7.
Datetime	1753-01-01 00:00:00.000	9999-12-31 23:59:59.997	8 bytes	Datetime	If you are running SQL Server 2008 or later and need milliseconds precision, use datetime2(3) instead to save 1 byte.

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Smalldatetime	1900-01-01 00:00	2079-06-06 23:59		Datetime	
Date	0001-01-01	9999-12-31	3 bytes	Datetime	
Time	00:00:00.0000000	23:59:59.9999999		Datetime	Specifying the precision is possible. TIME(3) will have milliseconds precision. TIME(7) is the highest and the default precision. Casting values to a lower precision will round the value.
Datetime2	0001-01-01 00:00:00.0000000	9999-12-31 23:59:59.9999999	Precision 1-2 = 6 bytes precision 3-4 = 7 bytes precision 5-7 = 8 bytes	Datetime	Combines the date datatype and the time datatype into one. The precision logic is the same as for the time datatype.
Datetimeoffset	0001-01-01 00:00:00.0000000 -14:00	9999-12-31 23:59:59.9999999 +14:00	Precision 1-2 = 8 bytes precision 3-4 = 9 bytes precision 5-7 = 10 bytes	Datetime	Is a datetime2 datatype with the UTC offset appended.
Char	0 chars	8000 chars	Defined width	String	Fixed width
Varchar	0 chars	8000 chars	2 bytes + number of chars	String	Variable width
Varchar(max)	0 chars	2 ³¹ chars	2 bytes + number of chars	String	Variable width
Text	0 chars	2,147,483,647 chars	4 bytes + number of chars	String	Variable width
Nchar	0 chars	4000 chars	Defined width x 2	Unicode	Fixed width
Nvarchar	0 chars	4000 chars		Unicode	Variable width
Nvarchar(max)	0 chars	2 ³⁰ chars		Unicode	Variable width
Ntext	0 chars	1,073,741,823 chars		Unicode	Variable width

DATATYPE	MIN	MAX	STORAGE	TYPE	NOTES
Binary	0 bytes	8000 bytes		Binary	Fixed width
Varbinary	0 bytes	8000 bytes		Binary	Variable width
Varbinary(max)	0 bytes	2 ³¹ bytes		Binary	Variable width
Image	0 bytes	2,147,483,647 bytes		Binary	Variable width. Prefer to use the varbinary(max) type as the image type will be removed in future versions.
Sql_variant				Other	Stores values of various SQL Server-supported data types, except text, ntext, and timestamp.
Timestamp				Other	Stores a database-wide unique number that gets updated every time a row gets updated.
Uniqueidentifier				Other	Stores a globally unique identifier (GUID).
Xml				Other	Stores XML data. You can store xml instances in a column or a variable.
Cursor				Other	A reference to a cursor.
Table				Other	Stores a result set for later processing.